CHANDERNAGORE COLLEGE

SEMESTER – IV B.Sc. (MAJOR) INTERNAL ASSESSMENT EXAMINATION, 2025

SUBJECT: COMPUTER SCIENCE COURSE CODE: COMP 4011 COURSE NAME: OPERATING SYSTEMS

FULL MARKS: 10 TIME: 30 MINUTES

ATTEMPT ANY 02 OUT OF THE FOLLOWING 03 QUESTIONS 2 x 5 = 10

Q1: What is PCB? Describe different Process States. What is Context Switch? [2 + 2 + 1]

Q2: Find the average waiting time of the processes if we use SRTF algorithm. [3 + 1 + 1]

Process	CPU Burst time (ms)	Arrival Time
P1	3	0
P2	6	2
P3	4	4
P4	5	6
P5	2	8

What is 'Aging'? What will happen if the 'Time Quantum' is too large in Round Robin Algorithm?

Q3: Determine the Number of Page Faults and Page fault rate for the reference string – 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1 with 3 available frames in 'Optimal Page Replacement' Algorithm. What is 'Belady's Anomaly'? [3 + 2]

Q4: What is the difference between 'Deadlock' and 'Livelock'? Write the Bankers 'Safety Algorithm' with an Example. What is its time complexity? [1 + 3 + 1]